Manual

Master Data Transfer from Third-Party Systems

EIS-SDF 8.1

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Contents

1 Master Data Transfer from Third-Party Systems

Summary

Fields of application

When introducing the MES system, different master data need to be created. In some cases, very large amounts of data need to be created. These data sets cannot be edited manually or it requires lots of maintenance work.

For this reason, it is possible to transfer master data automatically from third-party systems.

Implementation notes

You use the master data import function if you require to initially transfer large amounts of master data from external systems when introducing a new system.

Features

- Master data transfer from third-party systems
    - Function to transfer master data (checked for validity) from third-party systems including documentation to describe the data format in use and the steps required for posting
- Import of machine master data
- Import of resources
- Import of production variants
- Import of material buffers

Procedure

- Subject to the data type to be imported, the below-described procedure is to be considered in addition to the data descriptions:

Import data type	**Document**
DNC resources	transfer of DNC data

2 Structure of the DLG Format

Basics of HYDRA BAPI

Data is always posted to the database in accordance with basic guidelines ensuring their consistency and uniformity. This is why any writing access to the database is performed by programs providing a uniform interface to this end.

This means that all writing accesses to the HYDRA database irrespective of whether these are called via HYDRA applications or external applications/ systems are executed by a program with a defined interface.

This is mainly the HYDRA BAPI. It is used in the course of the master data transfer in order to transfer and post data provided by and processed in external systems to HYDRA.

BAPIs and dialog commands

Essentially, there is for each object (that can be maintained using the MOC) such a BAPI in HYDRA. Objects in this sense may be (production) orders or master data records. There are always different methods to access such an object. In the easiest case this is a method to create (INSERT), modify (UPDATE) or delete data records (DELETE).

In more complex cases and/or when this is requested by the application, also different methods are implemented. This may be modifying methods comprising an insertion or modification or additional application-specific methods.

Such a BAPI is called by a so called dialog command. This command is comprised of:

<Object>**.**<Method>

This is an exemplary (and incomplete) overview of the available objects and their selected methods

Object	**Methods**	**Comment**
ANR	INSERT UPDATE DELETE MODIFY	The ANR object designates the order.

Object	Methods	Comment
MNR	INSERT UPDATE DELETE	The MNR object designates machines/ workplaces.
FERTVAR	INSERT UPDATE DELETE	The FERTVAR object designates production variants.
RES	INSERT UPDATE DELETE	The RES object designates the resources of the module WRM and DNC.

Dialog data strings

After the initial BAPI call using the command, the use data will be transferred in a so-called dialog string or dialog data string. The use data in a dialog string are clearly identified by indicators, also designated as acronyms.

Such an acronym may represent at least one database field or also have controlling effects on postings. The acronym is always followed by the equal sign "=" and the value transferred for this acronym. The individual acronyms and their values are separated by pipes "|" from each other and from the dialog command.

Example:

```
DLG=FERTVAR.INSERT|FERTVAR.ATK=BLOO01052225000000|FERTVAR.MGRP=BW2000|
FERTVAR.RESTYP=WNR|FERTVAR.RES=BLOO01052225000000 2|
FERTVAR.SZY=17143|FERTVAR.TLG=2|
FERTVAR.BEM=BLOO01052225000000\|rose\|2\|rose\|BW2000|
FERTVAR.VER=1|FERTVAR.STA=F|FERTVAR.FIR:ATK=0|
```

Data formats/ mandatory acronyms

The descriptions of the acronyms are based on the following data types:

Type	Description
CHAR x	For the data type CHAR the information will be aligned to the left; unnecessary positions will be filled with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. For the NUMC data type only digits are allowed (ASCII-digits 30 hex to 39 hex). The numbers will be aligned to the right and unnecessary positions will be filled with zeros. Example: "00000002"
DEC x.y	Numeric field of the length x contains y decimal places. A data field in the HYDRA format is **preceded** by a sign ("+" or "-") and it contains a decimal point. Empty places must be filled with zeros. e.g. DEC 13.3: **-**1234567890.123

Each BAPI call ***must*** contain the following header data in the dialog data

<table>
<tr><th>Identification</th><th>Content</th><th colspan="2">Description</th></tr>
<tr><td>DLG</td><td>{BAPI call}</td><td colspan="2">Dialog identification: This dialog identification indicates the desired BAPI call</td></tr>
<tr><td rowspan="5">USR</td><td rowspan="5">NUM 4</td><td colspan="2">HYDRA user: This Hydra user number uniquely identifies a HYDRA client:</td></tr>
<tr><td>MOC:</td><td>USR = 20000 + MOC number
USR = 20000 + MOC</td></tr>
<tr><td>LAN terminal (LANT)
FB terminal (FBT):</td><td>USR = 2000 + terminal number
USR = 2000 + TNR</td></tr>
<tr><td>External terminals</td><td>USR = 3000 ... 3999</td></tr>
<tr><td>MLE-MDM</td><td>USR=9999</td></tr>
<tr><td>DAT</td><td>{mm/dd/yyyy}</td><td colspan="2">Date: current date in the format mm/dd/yyyy
"Today" can be used as placeholder for the dynamic determination.</td></tr>
<tr><td>ZEI</td><td>{seconds}</td><td colspan="2">Time: current time in the seconds format
"Now" can be used as placeholder for the dynamic determination.</td></tr>
</table>

Depending on the BAPI call, additional identifications must/ may be entered.

Data objects with files

Only the file names will be indicated in the dialog string for such objects that contain files in addition to the data fields of dialog data strings, e.g. document resources or DNC resources. The files themselves will be stored to defined data areas. The data import consists of two steps: Dialog data strings and files.

Dialog data strings - acronyms

The acronym to indicate the file is a field of the field type CHAR 128 that includes the file name. In most of the cases the name is only indicated without path - please see the documentation for the BAPI concerned.

Example: RES.SPEICHORT:DATA includes the file name without path and without extension of the attached DNC file. The storage location and the extension are defined before in the system via the resource type.

File format:

The file format is not important for the storage in HYDRA. The file will be stored to the specified storage location. The application will then interpret this file. For the import of master data it must be taken into account that the file must be stored to the directory specified for the application.

Example DNC files: The DNC type defines in which folder the files are and how they must be stored and interpreted.

Multilingual database contents

As part of SIS-HLM, there is now the possibility to define descriptive texts in several languages for specific objects in the database. Provided that this function is enabled on the system, these columns may generally also be filled by using the master data import. Please note the following:

- Specify the target language
  The target language can be transferred as additional acronym in the dialog data strings.
  Example:
  Machine master data is to be transferred. English (EN) is defined with language index 2 in the system. The dialog data string to transfer this data has to be structured as follows:

  DLG=MNR.INSERT|…|MNR.MNR=<Machine>|MNR.BEZK=English description|…|LANG=2|…

- Only one language can be transferred every time an import is started.

This means, that two or more import runs might be required, subject to the number of configured languages. Please note the following:

- The first import has to be performed using the *.INSERT method.
  This rule can be ignored if there is a method "*.MODIFY" for the object. As in this case, the system decides whether an INSERT or an UPDATE is to be performed.
- All other imports need to be performed by way of the method "*.UPDATE" indicating all key fields pertaining to the object, the language-dependent description and the target language using the acronym "LANG=n".
- If the system uses a separately generated, internal key for an object, this one has to be determined after the initial creation. This internal key then needs to be provided for the updates that follow.

3 Data Transfer via Excel

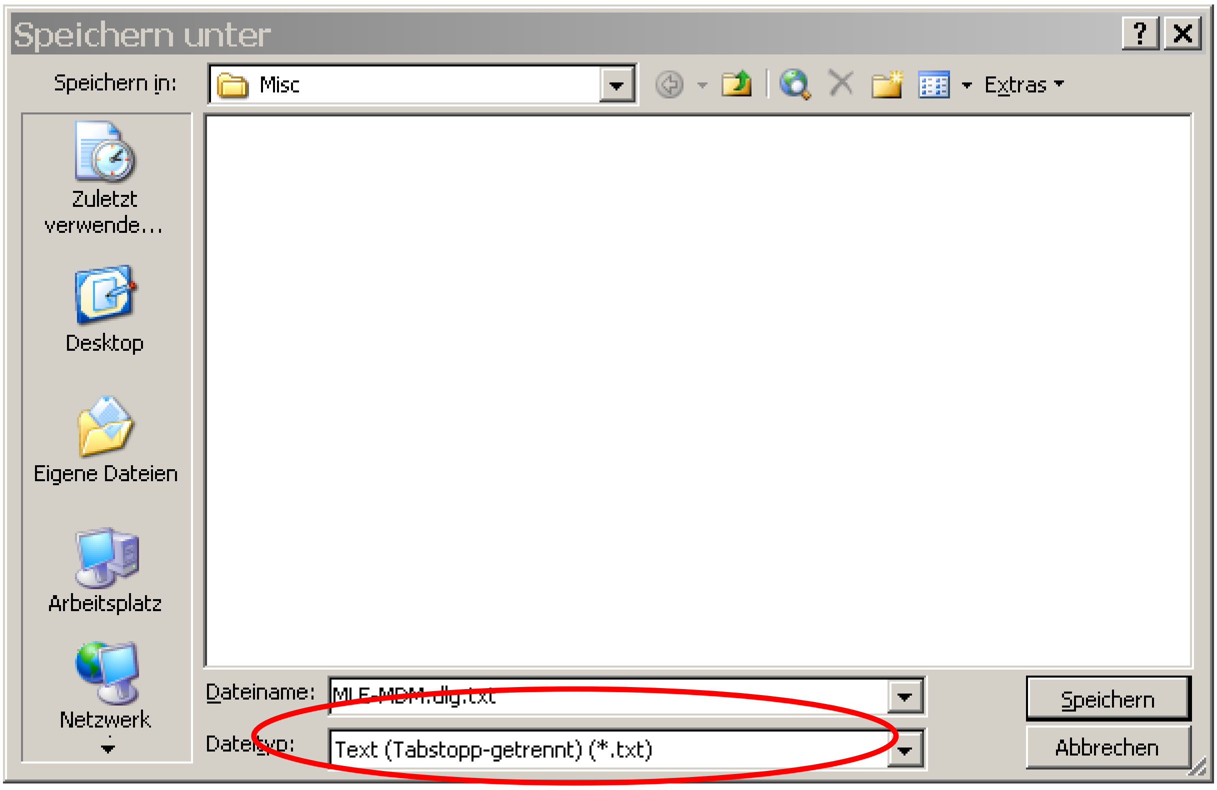
Experience has taught that preparing data in Microsoft Excel™ delivers the best results. But in general data may also be prepared in any other program. The paragraphs that follow show two possibilities of data provision.

Data provision in Excel (file *.xls)

MPDV provides a sample file to prepare data in Excel. Selected master data have already been prepared in this file.

The Excel file provides a separate spreadsheet for each master data object indicating the real name of this object. The respective spreadsheets already include the available acronyms. The columns, which the user has to fill out with user data, are highlighted in yellow. The corresponding documents describe the respective meaning of acronyms.

Every single spreadsheet has to be saved as “text (tab delimited) (*.txt)” to be able to transfer the data from the Excel worksheet into the HYDRA DLG format after completion.

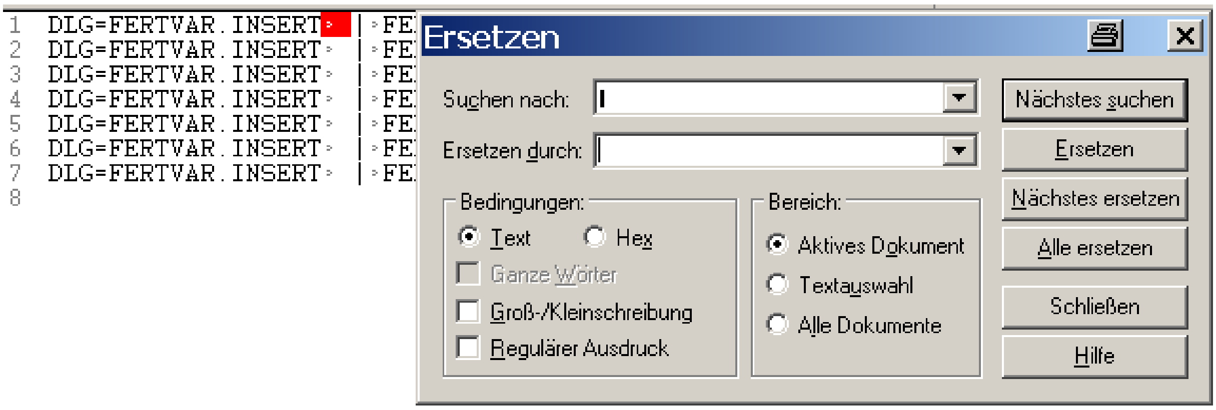


The resulting text file (MLE-MDM.DLG in the above example) has to be reworked in a text editor. In this case, the tabs included in the file have to be removed. This can be made in any text editor, unless it provides this feature. The TextPad (www.textpad.com) program has proved its worth in this connection. The procedure is described in an example on the basis of this program.

Once the file has been saved in the above-mentioned format in Excel, it is available as text file including tabs.

```
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
DLG=FERTVAR.INSERT   |FERTVAR.ATK=    |FERTVAR.MGRP=   |FERTVAR.RES
```

These tabs may now be removed by the “search/replace” function. To do so, select a tab, start the search/replace function and replace the tab by ““ (nothing).



Now we have a file in the HYDRA DLG format that can be imported in HYDRA.

```
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
```

Data provision as text file (file *.DLG

A file in the HYDRA DLG format needs to be generated anyway, even if Excel is not used. Such a file has the following, exemplary structure:

```
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
DLG=FERTVAR.INSERT|FERTVAR.ATK=|FERTVAR.MGRP=|FERTVAR.R
```

File import using the HYDRA server

The file can also directly be imported to HYDRA on the HYDRA server. But the procedure is slightly different for Windows and Unix.

File import with Windows

Connect to the HYDRA server (e.g. via RemoteDesktop). Start the Dos box from the HYDRA administration folder on the desktop. Please choose the Dos box for the correct system, if a HYDRA multi-system installation is in use.

Start the posting program in the Dos box as follows:

Hymwb.exe –d –u9999 –b<file name> > <file name>.pro

In case of the example from section 3:

Hymwb.exe –d –u9999 –bMLE-MDM.dlg > MLE-MDM.pro

The program is started by the parameter “-d” with developer traces. They ease the diagnosis if errors occur during the import. By entering the addition “> dlg.pro” the output is redirected into a log file, which simplifies checking at a later point in time.

Please note: The import only works properly if the posting program is run in the HYDRA directory. The file to be imported may be stored in any directory, the corresponding path has to be indicated in the parameter “-b”, e.g. „\importdir\datei.dlg“.

File import with Unix

Connect to the HYDRA server (e.g. via Telnet). Please choose the correct system if a HYDRA multi-system installation is in use.

Start the posting program as follows:

hymwb.out –d –u9999 –b<file name (case sensitive!)> > DLG.pro

In case of the example from section 3:

hymwb.out –d –u9999 –bMLE-MDM.DLG > MLE-MDM.pro

The program is started by the parameter “-d” with developer traces. They ease the diagnosis if errors occur during the import. By entering the addition “> dlg.pro” the output is redirected into a log file, which simplifies checking at a later point in time.

Please note: The import only works properly if the posting program is run in the HYDRA directory. The file to be imported may be stored in any directory. The corresponding path has to be indicated in the parameter “-b”, e.g. „\importdir\datei.dlg“.

Import of object files

In case objects include files (e.g. DNC), the files are copied to the target folder using the operating system functions. This can happen prior to or after the BAPI process. But file management and processing have to be configured beforehand. This depends on the application and is explained in the corresponding documents dealing with the application (e.g. the resource type DNC including the correct path configuration has to be created and configured).

4 Data type definitions

	→	

Type	Description
CHAR x	The information is left-aligned for the data type CHAR; unnecessary places are filled with blanks (blanks - (U+0020)). If a field is not used, fill it in full length with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. The NUMC data type only supports digits (ASCII characters 30 hex to 39 hex and/or U+0030 – U+0039). Numbers are right-aligned; unnecessary places are filled with zeros (U+0030). If a field is not used, fill it in full length with zeros. Example: "00000002"
DEC x.y **QUAN x.y**	Numeric field of the length x and y decimal places. A data field in HYDRA format is **preceded** by a sign ("+" or "-") and includes a decimal point. Enter zeros to fill empty places. If a field is not used, fill it in full length with zeros (U+0030) including sign and decimal separator. e.g. DEC 13,3: • `-1234567890,123` → `-1234567890.123` • `234567890,3` → `+0234567890.300` Note: The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. For example: a QUAN 13.3 field results in an external length of CHAR15.
DATE	Dates must be transferred in the HYDRA format **MM/DD/YYYY.** Populate unused date fields with blanks (U+0020; zero(s) (U+0030) not accepted).
TIME	Times must be transferred in the HYDRA format **seconds after midnight** (0 - 86400).

For all alphanumeric fields, HYDRA does not support specific special characters. These characters are: "****" (backslash - U+005B), "|" (pipe - U+007C), „ **"** " (double quote - U+0022), and „ **'** " (single quote - U+0027). You cannot enter these characters using the shop floor terminals;

the terminals and the MOC do not support these characters.

The character " **;** " (semicolon - U+003B) is used as separator for data collection. You must not use this character in key fields (e.g. order, batch number, personnel number, etc.).

The character " **%** " (percent - U+0025) is used as placeholder/wildcard character for database queries. For this reason, you should avoid using this character as it might falsify results.

In general, you must not use special characters ranging from U+0000 to U+001F. Exception: U+000A and U+000D as end-of-line characters.

The file must not include Byte Order Mark (BOM).

In general, HYDRA always expects a contiguous data structure. Consequently, you have to populate unused data fields with such default values that comply with the applicable conventions. This also applies to fields that are not required at the end of a data structure. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) und 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, you can also transfer files in the file format that was used until MW 2.x.

5 Material Buffer

Available methods

Method	Usage
MATPUF.INSERT	Create material buffer
MATPUF.UPDATE	Change material buffer
MATPUF.DELETE	Delete material buffer

Data

Column	Description
Field	Field name
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be completed.
T(type)	Data type of the field
L(ength)	Field length For fields of data type DEC: Overall number of digits without decimal separator and algebraic sign
D(ecimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant
Description	Description and/or comment of the field

Field	V	T	L	D	Description
MATPUF.MATPUF	S	CHAR	12		Material buffer
MATPUF.TYP		CHAR	1		Type (see available values in GUI)
MATPUF.BEZ		CHAR	30		Name
MATPUF.LAGORT		CHAR	20		Storage location
MATPUF.KST		CHAR	10		Cost center
MATPUF.ABT		CHAR	10		Department
MATPUF.BER		CHAR	10		Area
MATPUF.FIR		CHAR	4		Company
MATPUF.BEM		CHAR	20		Comment
MATPUF.DAUER		NUMC	7		Retention period
MATPUF.OPT:TANRPRN		CHAR	1		Internal use - do not transfer acronym
MATPUF.OPT:NOTMATPUF		CHAR	1		Internal use - do not transfer acronym
MATPUF.OPT:PKORB		CHAR	1		ID "recycle bin" DLG "J“ yes "N“ no
MATPUF.OPT:INBESTVER B		CHAR			ID "Include in stock" "J“ yes "N“ no
MATPUF.HARCID		NUMC	3		Hierarchy
MATPUF.HARCMATPUF		CHAR	12		Superordinate material buffer
MATPUF.ART		CHAR	1		Type of batch transport: "K” no buffer "E" input buffer "A" output buffer

Field	V	T	L	D	Description
MATPUF.ZLO		CHAR	10		Batch transport – corr. system
MATPUF.OPT:LAGVERB		CHAR	1		Internal use - do not transfer acronym
MATPUF.OPT:VIRTLAG		CHAR	1		ID "Virt. stock buffer" "J“ yes "N“ no

6 Resources

Available methods

Method	Usage
RES.INSERT	Create resource
RES.UPDATE	Change resource
RES.DELETE	Delete resource

Data

Column	Description
Field	Field name
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be completed.
T(ype)	Data type of the field
L(ength)	Field length For fields of data type DEC: Overall number of digits without decimal separator and algebraic sign
D(ecimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant
Description	Description and/or comment of the field

Field	V	T	L	D	Description
RES.RES	S	CHAR	20		Resource
RES.RESTYP	S	CHAR	8		Resource type
RES.BEZ		CHAR	40		Name
RES.VAB		CHAR	15		Responsibility area
RES.KST		CHAR	10		Cost center
RES.INVNR		CHAR	40		Inventory number
RES.GRAVNR		CHAR	40		Engraving number
RES.ZEICHNR		CHAR	20		Drawing number
RES.HERST		CHAR	40		Manufacturer
RES.EIGENT		CHAR	40		Owner
RES.ANSCHAFFDAT		DATE			Date of purchase
RES.ANSCHAFFKOST		DEC	9	2	Acquisition costs
RES.MATPUF:S		CHAR	10		Storage location
RES.LIEFDAT		DATE			Delivery date
RES.INBDAT		DATE			Start-up date
RES.GARDAT		DATE			Guarantee date
RES.BEZFREMD		CHAR	50		External name
RES.TYPBEZ		CHAR	50		Description of resource type
RES.VERW		CHAR	25		Use (see possible entries displayed in the GUI)
RES.BESTNR		CHAR	25		Purchase order number
RES.LIEFNR		CHAR	25		Supplier no.

Field	V	T	L	D	Description
RES.VERANT:TYP		CHAR	25		Party in charge Type (see possible entries displayed in the GUI)
RES.VERANT:NR		CHAR	25		Party in charge
RES.ANONYM		CHAR	1		Type "J“ Anonymous resource "N“ No anonymous resource "B“ Required resource
RES.OPT:TYPGL		CHAR	1		Equal type (see possible entries displayed in the GUI)
RES.RESVER		CHAR	12		Version
RES.ANZ		NUMC	5		Number
RES.RESFAMID		NUMC	7		Family
RES.SGR:HUB		NUMC	12		Cycles
RES.OPT:EINH		CHAR	3		Input unit
RES.SGR:KLKLZ		NUMC	9		Run time (in seconds)
RES.SZY		NUMC	9		Target cycle (in seconds/1000 cycles)
RES.TLG:S		NUMC	5		Original partitioning
RES.TLG:I		NUMC	5		Current partitioning
RES.OPT:AUTOANMELD		CHAR	1		Log on with OP "J“ log on resource with order when A_AN or log off resource with order when A_AB "N“ do not log on/off resource with order (if DNC always "N") "E“ explicit logon / change to logon allowed (as of version WRM 7.2)
RES.OPT:MULTIMNR		CHAR	1		ID "Can be logged on at the same time" "J“ yes "N“ no
RES.OPT:VERB		CHAR	1		ID "Post to resource" "J“ yes "N“ no
RES.ANFZ		NUMC	9		Setup time (in seconds)
RES.ABRZ		NUMC	9		Retooling (teardown) time (in seconds)
RES.OPT:BEL		CHAR	1		Assignment (see possible entries displayed in the GUI)
RES.OPT:AUSWSIB		CHAR	1		ID "Consider in evaluations" "J“ yes "N“ no
RES.SPEICHORT:DATA		CHAR	128		File name
RES.RES:V1		CHAR	20		Resource 1
RES.RESTYP:V1		CHAR	8		Resource type 1
RES.RES:V2		CHAR	20		Resource 2
RES.RESTYP:V2		CHAR	8		Resource type 2
RES.GENAUSKL		CHAR	50		Accuracy class
RES.EINHEIT		CHAR	3		Unit
RES.MESSBAB		DEC	10	4	Measurement range from
RES.MESSBBIS		DEC	10	4	Measurement range to
RES.MEISTM		DEC	10	4	Master value
RES.MEISTLAB		DEC	10	4	Master tolerance from
RES.MEISTLBIS		DEC	10	4	Master tolerance to
RES.USRFLD		CHAR	8		User field key
RES.FU:1		DATE	10		User field 1
RES.FU:2		DATE	10		User field 2
RES.FU:3		DATE	10		User field 3
RES.FU:4		DATE	10		User field 4

Field	V	T	L	D	Description
RES.FU:5		DATE	10		User field 5
RES.FU:6		DATE	10		User field 6
RES.FU:7		NUM	8		User field 7
RES.FU:8		NUM	8		User field 8
RES.FU:9		NUM	8		User field 9
RES.FU:10		NUM	8		User field 10
RES.FU:11		NUM	8		User field 11
RES.FU:12		NUM	8		User field 12
RES.FU:13		NUM	8		User field 13
RES.FU:14		NUM	8		User field 14
RES.FU:15		NUM	8		User field 15
RES.FU:16		NUM	8		User field 16
RES.FU:17		NUM	8		User field 17
RES.FU:18		NUM	8		User field 18
RES.FU:19		NUM	8		User field 19
RES.FU:20		NUM	8		User field 20
RES.FU:21		NUM	8		User field 21
RES.FU:22		NUM	8		User field 22
RES.FU:23		DEC	13	3	User field 23
RES.FU:24		DEC	13	3	User field 24
RES.FU:25		DEC	13	3	User field 25
RES.FU:26		DEC	13	3	User field 26
RES.FU:27		DEC	13	3	User field 27
RES.FU:28		DEC	13	3	User field 28
RES.FU:29		CHAR	1		User field 29
RES.FU:30		CHAR	1		User field 30
RES.FU:31		CHAR	1		User field 31
RES.FU:32		CHAR	1		User field 32
RES.FU:33		CHAR	1		User field 33
RES.FU:34		CHAR	1		User field 34
RES.FU:35		CHAR	1		User field 35
RES.FU:36		CHAR	1		User field 36
RES.FU:37		CHAR	1		User field 37
RES.FU:38		CHAR	1		User field 38
RES.FU:39		CHAR	1		User field 39
RES.FU:40		CHAR	1		User field 40
RES.FU:41		CHAR	1		User field 41
RES.FU:42		CHAR	1		User field 42
RES.FU:43		CHAR	1		User field 43
RES.FU:44		CHAR	1		User field 44
RES.FU:45		CHAR	10		User field 45
RES.FU:46		CHAR	10		User field 46
RES.FU:47		CHAR	10		User field 47
RES.FU:48		CHAR	10		User field 48
RES.FU:49		CHAR	10		User field 49
RES.FU:50		CHAR	10		User field 50
RES.FU:51		CHAR	20		User field 51
RES.FU:52		CHAR	20		User field 52

Field	V	T	L	D	Description
RES.FU:53		CHAR	20		User field 53
RES.FU:54		CHAR	20		User field 54
RES.FU:55		CHAR	20		User field 55
RES.FU:56		CHAR	20		User field 56
RES.FU:57		CHAR	20		User field 57
RES.FU:58		CHAR	20		User field 58
RES.FU:59		CHAR	20		User field 59
RES.FU:60		CHAR	20		User field 60
RES.FU:61		CHAR	20		User field 61
RES.FU:62		CHAR	20		User field 62
RES.FU:63		CHAR	20		User field 63
RES.FU:64		CHAR	20		User field 64
RES.FU:65		CHAR	40		User field 65
RES.FU:66		CHAR	40		User field 66
RES.BEM:1		CHAR	60		Comment field 1
RES.BEM:2		CHAR	60		Comment field 2
RES.BEM:3		CHAR	60		Comment field 3
RES.BEM:4		CHAR	60		Comment field 4
RES.BEM:5		CHAR	60		Comment field 5
RES.BEM:6		CHAR	60		Comment field 6

7 Production Variants/Methods

Available methods

Method	Usage
FERTVAR.INSERT	Create a production variant/method
FERTVAR.UPDATE	Change a production variant/method
FERTVAR.DELETE	Delete a production variant/method

Data

Column	Description
Field	Field name
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be completed.
T(ype)	Data type of the field
L(ength)	Field length For fields of data type DEC: Overall number of digits without decimal separator and algebraic sign
D(ecimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant
Description	Description and/or comment of the field

Field	V	T	L	D	Description
FERTVAR.VER	S	CHAR	10		Version
FERTVAR.ATK	S	CHAR	40		Article
FERTVAR.ATK:BEZ		CHAR	40		Article name
FERTVAR.MNR	S	CHAR	8		Workplace
FERTVAR.MGRP	S	CHAR	8		Group
FERTVAR.FIR:ATK	S	CHAR	10		Company for article
FERTVAR.RESTYP		CHAR	8		Resource type
FERTVAR.WANZ		NUMC	9		Number of resources
FERTVAR.RES		CHAR	20		Resource
FERTVAR.PRIO		NUMC	9		Priority
FERTVAR.SZY		NUMC	9		Target cycle (in seconds/1000 cycles)
FERTVAR.SZY:ABW		DEC	5	2	Admissible deviation
FERTVAR.TLG		NUMC	9		Partitioning
FERTVAR.RUEZ		NUMC	9		Setup time
FERTVAR.ABRZ		NUMC	9		Teardown/retooling time
FERTVAR.DATB		DATE			Valid from
FERTVAR.DATE		DATE			Valid until
FERTVAR.STA		CHAR	1		Status "F" Released "S" Blocked
FERTVAR.GRTXTNR		NUMC	4		Blocking reason
FERTVAR.BEM		CHAR	40		Comment

Field	V	T	L	D	Description
FERTVAR.USRFLD		CHAR	8		User field key
FERTVAR.FU:1		DATE	10		User field 1
FERTVAR.FU:2		DATE	10		User field 2
FERTVAR.FU:3		DATE	10		User field 3
FERTVAR.FU:4		DATE	10		User field 4
FERTVAR.FU:5		DATE	10		User field 5
FERTVAR.FU:6		DATE	10		User field 6
FERTVAR.FU:7		NUM	8		User field 7
FERTVAR.FU:8		NUM	8		User field 8
FERTVAR.FU:9		NUM	8		User field 9
FERTVAR.FU:10		NUM	8		User field 10
FERTVAR.FU:11		NUM	8		User field 11
FERTVAR.FU:12		NUM	8		User field 12
FERTVAR.FU:13		NUM	8		User field 13
FERTVAR.FU:14		NUM	8		User field 14
FERTVAR.FU:15		NUM	8		User field 15
FERTVAR.FU:16		NUM	8		User field 16
FERTVAR.FU:17		NUM	8		User field 17
FERTVAR.FU:18		NUM	8		User field 18
FERTVAR.FU:19		NUM	8		User field 19
FERTVAR.FU:20		NUM	8		User field 20
FERTVAR.FU:21		NUM	8		User field 21
FERTVAR.FU:22		NUM	8		User field 22
FERTVAR.FU:23		DEC	13	3	User field 23
FERTVAR.FU:24		DEC	13	3	User field 24
FERTVAR.FU:25		DEC	13	3	User field 25
FERTVAR.FU:26		DEC	13	3	User field 26
FERTVAR.FU:27		DEC	13	3	User field 27
FERTVAR.FU:28		DEC	13	3	User field 28
FERTVAR.FU:29		CHAR	1		User field 29
FERTVAR.FU:30		CHAR	1		User field 30
FERTVAR.FU:31		CHAR	1		User field 31
FERTVAR.FU:32		CHAR	1		User field 32
FERTVAR.FU:33		CHAR	1		User field 33
FERTVAR.FU:34		CHAR	1		User field 34
FERTVAR.FU:35		CHAR	1		User field 35
FERTVAR.FU:36		CHAR	1		User field 36
FERTVAR.FU:37		CHAR	1		User field 37
FERTVAR.FU:38		CHAR	1		User field 38
FERTVAR.FU:39		CHAR	1		User field 39
FERTVAR.FU:40		CHAR	1		User field 40
FERTVAR.FU:41		CHAR	1		User field 41
FERTVAR.FU:42		CHAR	1		User field 42
FERTVAR.FU:43		CHAR	1		User field 43
FERTVAR.FU:44		CHAR	1		User field 44
FERTVAR.FU:45		CHAR	10		User field 45
FERTVAR.FU:46		CHAR	10		User field 46
FERTVAR.FU:47		CHAR	10		User field 47

Field	V	T	L	D	Description
FERTVAR.FU:48		CHAR	10		User field 48
FERTVAR.FU:49		CHAR	10		User field 49
FERTVAR.FU:50		CHAR	10		User field 50
FERTVAR.FU:51		CHAR	20		User field 51
FERTVAR.FU:52		CHAR	20		User field 52
FERTVAR.FU:53		CHAR	20		User field 53
FERTVAR.FU:54		CHAR	20		User field 54
FERTVAR.FU:55		CHAR	20		User field 55
FERTVAR.FU:56		CHAR	20		User field 56
FERTVAR.FU:57		CHAR	20		User field 57
FERTVAR.FU:58		CHAR	20		User field 58
FERTVAR.FU:59		CHAR	20		User field 59
FERTVAR.FU:60		CHAR	20		User field 60
FERTVAR.FU:61		CHAR	20		User field 61
FERTVAR.FU:62		CHAR	20		User field 62
FERTVAR.FU:63		CHAR	20		User field 63
FERTVAR.FU:64		CHAR	20		User field 64
FERTVAR.FU:65		CHAR	40		User field 65
FERTVAR.FU:66		CHAR	40		User field 66

Information on key fields:

The fields "group" (FERTVAR.MGRP) and "workplace" (FERTVAR.MNR) can be entered separately or together. But one of the two values must be entered in any case.

8 Machines / Workplaces

Available methods

Method	Usage
MNR.INSERT	Create machine/workplace
MNR.UPDATE	Change machine/workplace
MNR.DELETE	Delete machine/workplace

Data

Column	Description
Field	Field name
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be completed. M Mandatory field
T(ype)	Data type of the field
L(ength)	Field length For fields of data type DEC: Overall number of digits without decimal separator and algebraic sign
D(ecimal places)	For fields of data type DEC: Number of decimal places; otherwise: not relevant
Description	Description and/or comment of the field

Field	V	T	L	D	Description
BEARB	M	CHAR	10		HYDRA User
MNR.MNR	S	CHAR	8		Machine/workplace
MNR.BEZK		CHAR	8		Short name
MNR.BEZL		CHAR	40		Name
MNR.VAB		CHAR	15		Responsibility area
MNR.KST		CHAR	10		Cost center
MNR.TYP		CHAR	1		Workplace category (see possible values displayed in the GUI)
MNR.SPERR		CHAR	1		ID "blocked" "J“ yes "N“ no
MNR.ART		CHAR	1		Workplace type (see possible values displayed in the GUI)
MNR.OPT:FREMDAPZ		CHAR	1		ID "external workplace" "J“ yes "N“ no
MNR.FIR		CHAR	4		Company
MNR.MGRP		CHAR	8		Group
MNR.CAT		CHAR	10		Category
MNR.BDEJMOD		NUMC	3		Year model
MNR.MSTDSATZ		DEC	6	2	Standard rate machine
MNR.PSTDSATZ		DEC	6	2	Standard labor rate
MNR.LEIGRAD		NUMC	3		Performance level

Field	V	T	L	D	Description
MNR.AKKORD		CHAR	1		Incentive wage indicator (see possible values displayed in the GUI)
MNR.ICON		CHAR	20		File name
MNR.OPT:MULTIAG		CHAR	1		Logon of multiple OPs (see possible values displayed in the GUI)
MNR.OPT:VLISTMOD		CHAR	1		Sequencing list (see possible values displayed in the GUI)
MNR.VLISTANZ		NUMC	3		Number of OPs in sequencing list
MNR.OPT:VLISTZW		CHAR	1		Compulsory sequence (see possible values displayed in the GUI)
MNR.VISLIST3		CHAR	10		Display 3rd list The field includes none, one or several of these options. They are separated by semicolon: “M” Input material “R” Resources “P“ Staff "A“ Output material Example: "M;R;P;A" if all options are set
MNR.VISFHMTNRAAN		CHAR	1		Material/PRT list when logging on OPs (see possible values displayed in the GUI)
MNR.DLGSTRG		CHAR	10		Dialog control
MNR.OPT:MAABP		CHAR	1		Quantity posting to staff (see possible values displayed in the GUI)
MNR.OPT:AGIST		CHAR	1		ID "Posting on OPs not logged on" "J“ yes "N“ no
MNR.OPT:ANTDAUER		CHAR	1		Posting of machine time for simultaneous OPs (see possible values displayed in the GUI)
MNR.OPT:AANSKBAUTO		CHAR	1		Log OP on automatically when shift ends (see possible values displayed in the GUI)
MNR.OPT:PABSKE		CHAR	1		Log person off automatically when shift ends (see possible values displayed in the GUI)
MNR.PLANFKT		CHAR	1		Planning function (see possible values displayed in the GUI)
MNR.PLANJMOD		NUMC	3		Planned year model
MNR.KAPJMOD		NUMC	5		Availability (per mil)
MNR.OPT:CHV		CHAR	1		Batch management (see possible values displayed in the GUI)
MNR.MATPUF:IN		CHAR	12		Preceding material buffer
MNR.MATPUF:OUT		CHAR	12		Subsequent material buffer
MNR.OPT:CNRAUTOGEN		CHAR	1		Automatic generation of batch number (see possible values displayed in the GUI)
MNR.VISVERBRBLZ		CHAR	1		ID "Consumption balance" "J“ yes "N“ no
MNR.OPT:TRANROUT		CHAR	1		ID "Generate transport order for output material" "J“ yes "N“ no
MNR.OPT:TRANRIN		CHAR	1		ID "Generate transport order for input material" "J“ yes "N“ no
MNR.VERB:GUT		CHAR	3		Allocation of yield "AUS“ Scrap "NCH“ Rework "PRB“ Open quantity

Field	V	T	L	D	Description
MNR.VERB:AUS		CHAR	3		Allocation of scrap "GUT“ Yield "NCH“ Rework "PRB“ Open quantity
MNR.VERB:NCH		CHAR	3		Allocation of rework "GUT“ Yield "AUS“ Scrap "PRB“ Open quantity
MNR.VERB:PRB		CHAR	3		Allocation of open quantity "GUT“ Yield "AUS“ Scrap "NCH“ Rework
MNR.OPT:GUTMANU		CHAR	1		ID "Manual entry of yield" "J“ yes "N“ no
MNR.OPT:AUSMANU		CHAR	1		ID "Manual entry of scrap" "J“ yes "N“ no
MNR.OPT:NCHMANU		CHAR	1		ID "Manual entry of rework quantity" "J“ yes "N“ no
MNR.OPT:PRBMANU		CHAR	1		ID "Manual entry of open quantity" "J“ yes "N“ no
MNR.OPT:GUTMANUTAKT		CHAR	1		ID "Posting of yield as cycles" "J“ yes "N“ no
MNR.OPT:AUSMANUTAKT		CHAR	1		ID "Posting of scrap as cycles" "J“ yes "N“ no
MNR.OPT:NCHMANUTAKT		CHAR	1		ID "Posting of rework as cycles" "J“ yes "N“ no
MNR.OPT:PRBMANUTAKT		CHAR	1		ID "Posting of open quantity as cycles" "J“ yes "N“ no
MNR.OPT:UMRMENGE		CHAR	1		Basis for MDE quantity conversion (see possible values displayed in the GUI)
MNR.EGE:GUTP		CHAR	3		Quantity unit (P)
MNR.UMRFAKTP:Z		NUMC	9		Quantity unit (P) - numerator, primary quantity
MNR.UMRFAKTP:N		NUMC	9		Quantity unit (P) - denominator, primary quantity
MNR.EGE:GUTS		CHAR	3		Quantity unit (S)
MNR.UMRFAKTS:Z		NUMC	9		Quantity unit (S) - numerator, primary quantity
MNR.UMRFAKTS:N		NUMC	9		Quantity unit (S) - denominator, primary quantity
MNR.EGE:GUTT		CHAR	3		Quantity unit (T)
MNR.UMRFAKTT:Z		NUMC	9		Quantity unit (T) - numerator, primary quantity
MNR.UMRFAKTT:N		NUMC	9		Quantity unit (T) - denominator, primary quantity
MNR.EGE:GUTB		CHAR	3		Quantity unit (B)
MNR.UEBART		CHAR	1		Monitoring type “Z” Cyclic monitoring "B” Operating signal "K“ No monitoring
MNR.UEBDAUER		NUMC	3		Minimum cycle/malfunction time (seconds)

Field	V	T	L	D	Description
MNR.IZYABW		NUMC	4		Cycle extension
MNR.ANZSTAKT		NUMC	4		Number of target cycles
MNR.MWANZ		NUMC	1		Cycles to be evaluated
MNR.OPT:MSTAUAUF		CHAR	1		ID "Activation of required malfunction reason input" "J“ yes "N“ no
MNR.OPT:MSTAUDAUER		NUMC	4		Delay time (seconds)
MNR.BUCHPSPERRE		CHAR	1		Posting during prod. lock "G" Posting as yield "A” Posting as scrap "X” No posting
MNR.IMPFAKT		NUMC	3		Pulse factor specific to machines
MNR.TLG		NUMC	5		Partitioning specific to machines
MNR.STKZG		NUMC	4		Waiting period short-term malfunction (seconds)
MNR.OPT:WENDAUTO		CHAR	1		ID "Extended weekend automatic" "J“ yes "N“ no
MNR.DIGOUT:MSPERRE		NUMC	2		Output "machine lock"
MNR.DIGOUT:SMENGE		NUMC	2		Output "target quantity reached"
MNR.DIGOUT:STOER		NUMC	2		Output "machine down"
MNR.DIGIO		NUMC	2		Free I/O
MNR.DIGIN:CAWL		NUMC	2		Output batch change
MNR.OPT:PDV		CHAR	1		ID "Collect process data" "J“ yes "N“ no
MNR.EXTTYP		CHAR	1		External connection "K” No external connection "J” DS100 "N“ MT3 "E“ Engel interfacing "A“ Arburg control system "P" PDE (Process Data Collection)
MNR.EXTSNR		NUMC	8		Serial number
MNR.EXTID		NUCM	2		Device address
MNR.USRFLD		CHAR	8		User field key
MNR.FU:1		DATE	10		User field 1
MNR.FU:2		DATE	10		User field 2
MNR.FU:3		DATE	10		User field 3
MNR.FU:4		DATE	10		User field 4
MNR.FU:5		DATE	10		User field 5
MNR.FU:6		DATE	10		User field 6
MNR.FU:7		NUM	8		User field 7
MNR.FU:8		NUM	8		User field 8
MNR.FU:9		NUM	8		User field 9
MNR.FU:10		NUM	8		User field 10
MNR.FU:11		NUM	8		User field 11
MNR.FU:12		NUM	8		User field 12
MNR.FU:13		NUM	8		User field 13
MNR.FU:14		NUM	8		User field 14
MNR.FU:15		NUM	8		User field 15
MNR.FU:16		NUM	8		User field 16

Field	V	T	L	D	Description
MNR.FU:17		NUM	8		User field 17
MNR.FU:18		NUM	8		User field 18
MNR.FU:19		NUM	8		User field 19
MNR.FU:20		NUM	8		User field 20
MNR.FU:21		NUM	8		User field 21
MNR.FU:22		NUM	8		User field 22
MNR.FU:23		DEC	13	3	User field 23
MNR.FU:24		DEC	13	3	User field 24
MNR.FU:25		DEC	13	3	User field 25
MNR.FU:26		DEC	13	3	User field 26
MNR.FU:27		DEC	13	3	User field 27
MNR.FU:28		DEC	13	3	User field 28
MNR.FU:29		CHAR	1		User field 29
MNR.FU:30		CHAR	1		User field 30
MNR.FU:31		CHAR	1		User field 31
MNR.FU:32		CHAR	1		User field 32
MNR.FU:33		CHAR	1		User field 33
MNR.FU:34		CHAR	1		User field 34
MNR.FU:35		CHAR	1		User field 35
MNR.FU:36		CHAR	1		User field 36
MNR.FU:37		CHAR	1		User field 37
MNR.FU:38		CHAR	1		User field 38
MNR.FU:39		CHAR	1		User field 39
MNR.FU:40		CHAR	1		User field 40
MNR.FU:41		CHAR	1		User field 41
MNR.FU:42		CHAR	1		User field 42
MNR.FU:43		CHAR	1		User field 43
MNR.FU:44		CHAR	1		User field 44
MNR.FU:45		CHAR	10		User field 45
MNR.FU:46		CHAR	10		User field 46
MNR.FU:47		CHAR	10		User field 47
MNR.FU:48		CHAR	10		User field 48
MNR.FU:49		CHAR	10		User field 49
MNR.FU:50		CHAR	10		User field 50
MNR.FU:51		CHAR	20		User field 51
MNR.FU:52		CHAR	20		User field 52
MNR.FU:53		CHAR	20		User field 53
MNR.FU:54		CHAR	20		User field 54
MNR.FU:55		CHAR	20		User field 55
MNR.FU:56		CHAR	20		User field 56
MNR.FU:57		CHAR	20		User field 57
MNR.FU:58		CHAR	20		User field 58
MNR.FU:59		CHAR	20		User field 59
MNR.FU:60		CHAR	20		User field 60
MNR.FU:61		CHAR	20		User field 61
MNR.FU:62		CHAR	20		User field 62
MNR.FU:63		CHAR	20		User field 63
MNR.FU:64		CHAR	20		User field 64

Field	V	T	L	D	Description
MNR.FU:65		CHAR	40		User field 65
MNR.FU:66		CHAR	40		User field 66

The resource BAPI (RES.UPDATE) must be used to edit general information of the machine configuration, e.g. inventory no. engraving no., drawing no., manufacturer, owner, acquisition costs, supplier information and responsibilities.

9 Sample Files

Overview

Attached to this documentation, you will find test files for the interface EIS-SDF. The attachment is only available, if the documentation is in PDF format.

The documentation Opening attachments of a PDF document describes how to call the attached test files.

The following test files are attached to the PDF document:

File	**Comment**
machine_master_data.xlsx	Example that demonstrates how master data is transferred for workplaces.
WRM_productionvariants_resources.xls	Example that demonstrates how master data is transferred for production variants and resources.